

STUDENT COURSE MANAGEMENT SYSTEM

Project Report

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Course Title: COS 201 Introduction to Programming I

Project Title: Student Course Management System

Programming Language: Java

1. Introduction

The Student Course Management System is a Java console application developed to help users manage student course records efficiently. The system enables users to add new courses, display all available courses, search for a specific course using its course code, compute the total number of course units, save course records to a file, and load previously saved records from a file.

The application was developed using Object-Oriented Programming (OOP) principles in Java and demonstrates the use of classes, objects, arrays (ArrayList), file handling, exception handling, recursion, loops, and decision-making structures.

2. Objectives

The objectives of this project are to:

- Develop a menu-driven Java application.
- Apply Object-Oriented Programming concepts.
- Store and manage course information.
- Demonstrate file input and output operations.
- Implement recursion for searching courses.
- Handle invalid user inputs using exception handling.

3. System Features

The Student Course Management System provides the following features:

Add Course

The system allows users to enter a course code, course title, and course unit. Before adding a course, it checks whether another course with the same course code already exists. Duplicate course codes are not allowed.

View All Courses

This feature displays all recorded courses in a well-formatted table showing the course code, course title, and course units.

Search Course

The user can search for a course by entering its course code. The search operation is implemented using a recursive method, which searches through the list until the course is found or the end of the list is reached.

Compute Total Units

The program calculates the total number of units for all recorded courses using a loop that sums the units of every course stored in the system.

Save to File

All course records can be saved into a data file named **Course.dat**. Each course is stored in comma-separated format, making it easy to reload later.

Load from File

Previously saved records are loaded from **Course.dat** whenever requested or when the application starts. This allows course information to be retained even after the program is closed.

4. Programming Concepts Used

The project demonstrates several Java programming concepts, including:

- Classes and Objects
 - Constructors
 - Encapsulation using private variables and public getter methods
 - ArrayList for dynamic data storage
 - Loops (for-each loop)
 - Decision making using switch statements
 - Recursion for course searching
 - File handling using FileReader, FileWriter, BufferedReader, and PrintWriter
 - Exception handling using try-catch blocks
 - String manipulation using trim() and toUpperCase()
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5. Challenges Encountered

During the development of the project, several challenges were encountered.

Initially, organizing the project into multiple Java files caused compilation errors due to incorrect class separation and project structure. These issues were resolved by ensuring that each class was placed in its appropriate Java file and that the Main class contained the program entry point.

Another challenge was preventing duplicate course codes. This was solved by implementing a recursive search method that checks whether a course already exists before allowing a new course to be added.

File handling also required careful implementation to ensure that course records could be saved and loaded correctly without data loss.

6. Conclusion

The Student Course Management System successfully meets the project requirements. It provides an efficient way to manage course records while demonstrating important Java programming concepts such as Object-Oriented Programming, recursion, exception handling, loops, file handling, and decision-making structures.

The project improved my understanding of Java programming and strengthened my practical knowledge of software development techniques used in building console-based applications.

7. GitHub Repository

The complete source code for this project has been uploaded to GitHub to facilitate version control, project sharing, and future improvements. The repository contains the Java source files, project documentation, and supporting files required to run the application.

Repository URL: <https://github.com/ayenibernard/COS-201->